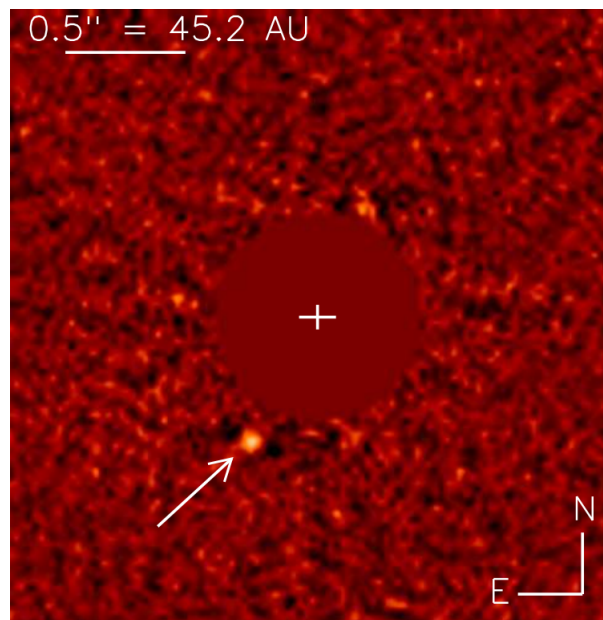

SUMMER RESEARCH 2018

Searching For Extrasolar Giant Planets with Direct Imaging

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November 19, 2018 - January 18, 2019



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1 INTRODUCTION

1.1 Background

Many different methods have been implemented to search for exoplanets, the common ones are transit and radial velocity, and other alternatives are through gravitational micro-lensing, direct imaging and astrometric wobbles. Among these techniques, direct imaging is one of the most challenging. Since planets do not usually emit light themselves, rather they reflect light from the host star, planets are always several magnitudes fainter than the star. Even if the planets are young and hence self-luminous, the emitted light is still much fainter compared to the star's emission. With that said, the contrast of a companion with respect to a star is extremely high and such contrast is very difficult to achieve through regular imaging techniques. The separation between the planet and the host star should also be taken into consideration. If the separation is too small, the current available instruments might not yet have the angular resolution to distinguish between the two point sources if they are too faraway from Earth. An analogy to these challenges, according to Dr. Valentin Christiaens, is to imagine trying to find a firefly in the vicinity of a bright lighthouse.

Despite the difficulties, it is still possible to image planets directly provided if we use the right technique and choose the source wisely. This 6-week research project specifically aims to find young, hot and self-luminous giant exoplanets, aged around 20-50 million years old, embedded in the debris disk around newly formed (young) stars through a technique called high contrast imaging. These adolescent planets are still believed to be emitting residual latent heat from their formation, hence the search is carried out using data collected by the Clio2 Near IR (L-band) camera of the MagAO adaptive optics (AO) system of the 6.5-m Magellan Clay telescope located at Las Campanas Observatory, Chile.^[1] For this project, I was provided with two sets of MagAO/Clio2 data: one of Gamma Doradus and another of HD35650. Both the targets have debris disks, indicating that the stars are relatively young and thus suspected to host at least one observable, young companion.

1.2 Project Overview

Before analysis can be carried out on the datasets, they first have to be reduced and calibrated. I was provided with the calibration pipeline for NaCo data but since the procedures in data calibration are instrument-dependent, the NaCo pipeline had to be adapted so that it is compatible to MagAO data. The next step is post-processing where Angular Differential Imaging (ADI) and Principal Component Analysis (PCA) are implemented on the calibrated data cube to produce final image of improved contrast, hence increasing the chances of finding a faint companion.

Although no companions were found in both Gamma Doradus and HD35650, the contrast curves computed from post-processing contains valuable information on the sensitivity of the instrument in detecting companions. The original contrast curves provides the flux sensitivity of the instrument, but it can be converted into mass sensitivity using evolutionary models of low-mass objects to predict the types of companions that can be seen through high contrast imaging.

In this report, the Calibration, Post-processing and Magnitude-to-mass conversion procedures will be elaborated in terms of Gamma Doradus, but the exact same procedures apply to HD35650 unless stated otherwise.

2 MagAO/Clio2 - GAMMA DORADUS RAW DATA

2.1 Data Cube Information

- (a) **Science target:** Gamma Doradus

- (b) **Target Description:** Gamma Doradus is the third brightest star in the Dorado constellation. This target is selected due to the presence of a debris disk which suggests that the star is relatively young and there could potentially be a self-emitting young companion.

- (c) **No. of Data Cubes:**
 - Total: 314
 - Science (saturated) cubes: 274
 - Unsaturated cubes: 40

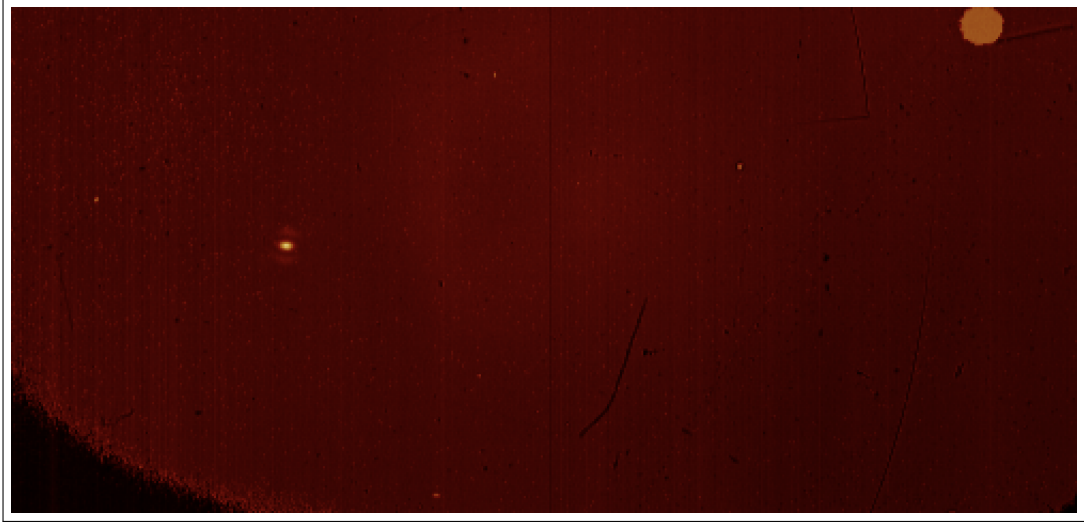
- (d) **No. of frames per cube:**
 - Science cube: 20
 - Unsaturated cube: 20

- (e) **Integration time per frame:**
 - Science frames: 280 ms
 - Unsaturated frames: 86 ms

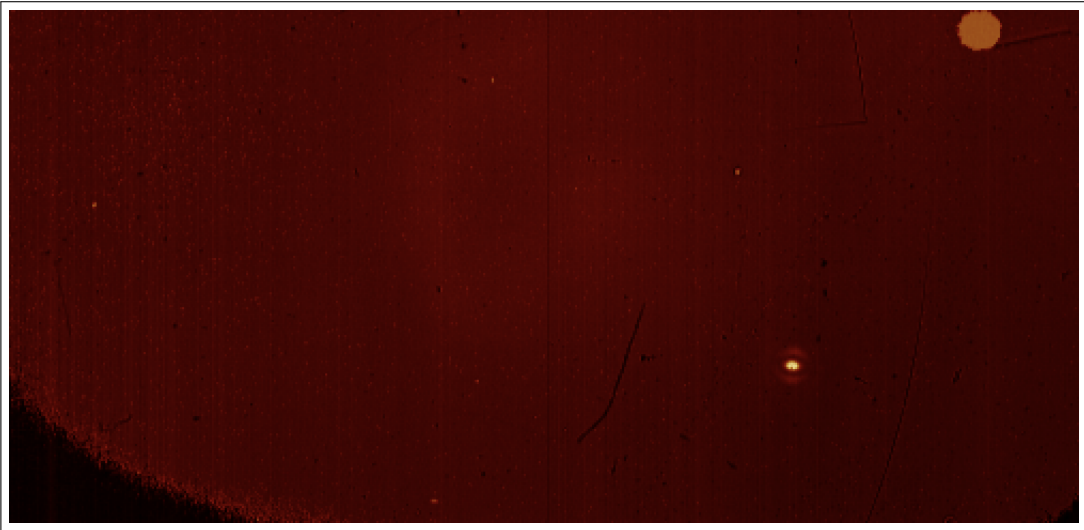
2.2 Telescope Dithering

Observations of target using MagAO/Clio2 are done using a process called dithering where the telescope shifts its field of view as the target is being observed. An example is shown in Figure 2.1 where the targets can be seen in both images but they appear at a different location on the detector.

For Gamma Doradus observation, the telescope dithers to a different configuration after every 10 cubes of image taken (i.e. telescope shifts every $10 \text{ cubes} \times 20 \text{ frames} \times 280 \text{ ms}$ from position A to B or vice versa). The dithering for this specific target per observation follows the order "ABBA" - meaning the telescope is shifted from position A to B to B to A. This means there will be 40 cubes in total in each complete observation of the target. The reason for performing the observation in this manner will be explained Section 3.3 - Sky Subtraction.



(a) Image of Gamma Doradus taken with telescope at position A.



(b) Image of Gamma Doradus taken with telescope at position B.

Figure 2.1

3 DATA CALIBRATION

3.1 Data Sorting

The first step before calibrating the data is to sort the files based on the configuration of the telescope during observation, and whether they are saturated or unsaturated. Since there are four dithering positions - A1, B1, B2, A2 for the ABBA dithering order of the telescope, there will be eight separate lists containing names of files:

- sci_A1 = [Gamma_doradus00001.fits,...]
- sci_B1
- sci_B2
- sci_A2
- unsat_A1
- unsat_B1
- unsat_B2
- unsat_A2

3.2 Linear Correction

When a detector is exposed to a bright source for a relatively long period of time, the photon counts will stop increasing as the detector has reached its photon counting capacity. This would lead to a departure from linearity in the relationship between photon count and detector integration time (DIT).

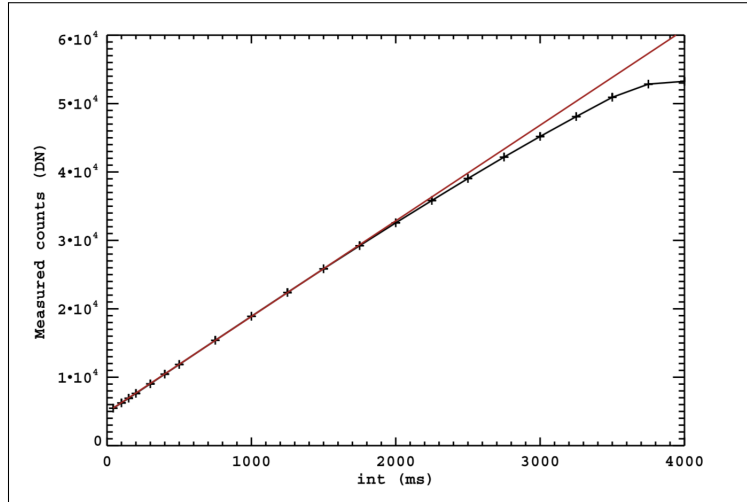


Figure 3.1: The red line is the ideal best-fit linear relationship between the measured counts and DIT. The black line with the x-marks is the raw linearity data which shows that the detector is strictly linear up to only 27,000 - 30,000 DN. (Morzinski et al. 2015)

To fix this problem, linear correction has to be performed on all the data cubes using a best-fit polynomial. In this case, a 3rd-order polynomial is selected (see Figure 3.2) and the linearity

correction formula which is to be applied to measured counts $> 27,000$ is:^[2]

$$y = 112.575 + 1.00273x - (1.40666 \times 10^{-6})x^2 + (4.59015 \times 10^{-11})x^3 \quad (1)$$

with y being the true count and x being the measured counts.

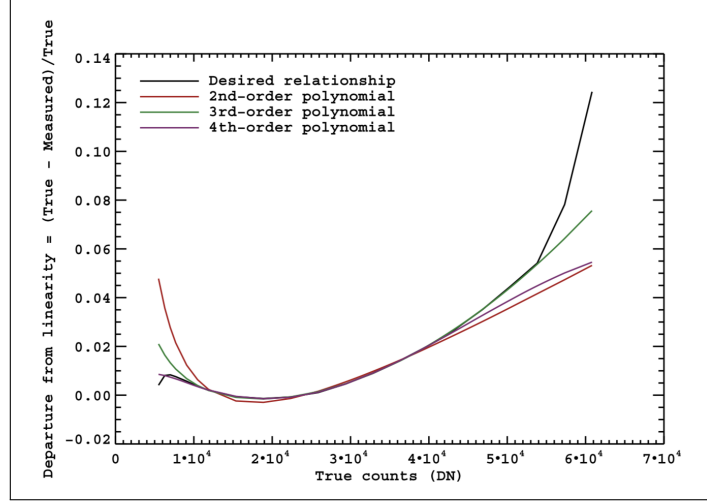


Figure 3.2: Here we can see that the 3rd-order polynomial is the best-fit to the desired relationship. (Morzinski et al. 2015)

Figure 3.3 shows the linear-corrected data using the polynomial equation given above. Note that only counts up to $\sim 45,000$ DN can be corrected for linearity and any raw counts above this threshold is avoided. Hence the data is only linearised up to $\sim 46,000$ DN.^[2]

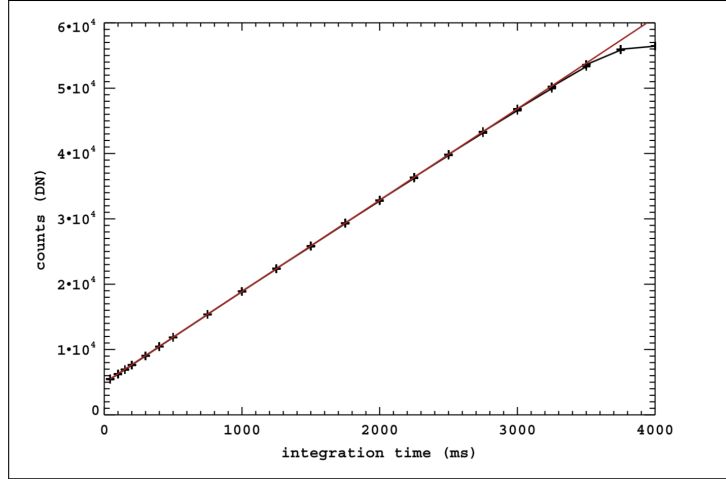


Figure 3.3: The red line is once again the best-fit linear relationship and the black line is the linear-corrected plot of the raw linearity data in Figure 3.1. (Morzinski et al. 2015)

3.3 Sky Subtraction

Usually in photometric calibration, flat field correction is an essential step to account for the non-uniformity of the detector's response to illumination due to variations in the pixel-to-pixel sensitivity of the detector. The flat field is taken by illuminating the detector with uniform source.

Unfortunately in the case of MagAO/Clio2, the flat field images of the instrument displayed a "glow" in the top centre of the image and cannot be removed even after many attempts using different techniques. The cause of this signature is still uncertain, but it is present in all the flats hence causing them to be unusable for flat field correction. (see Morzinski et al. 2015 for more details).

Since flat field correction cannot be done on the images of MagAO/Clio2, the next best thing that could be done is sky subtraction. As the term itself implies, we are trying to subtract the background sky away from the images. In this process, the imperfections in the instruments are also subtracted.

This procedure requires two images taken at roughly the same time in the night, with the science targets at different locations on the detector. This is why dithering of the telescope while observing is crucial for MagAO/Clio2 as previously elaborated in Section 2.2.

By taking images of the target with different telescope configuration allows us to subtract an image taken at position A from an image taken at position B and vice versa. This will leave us with a final image that will have approximately zero count everywhere except for where the science targets are in each frame of different telescopes. Though the science target will appear twice in the final image, one has positive flux and the other has a negative flux. The negative flux target will then be removed by cropping out the the frame which consists of the positive flux target. This is elaborated in more detail in Section 3.5.

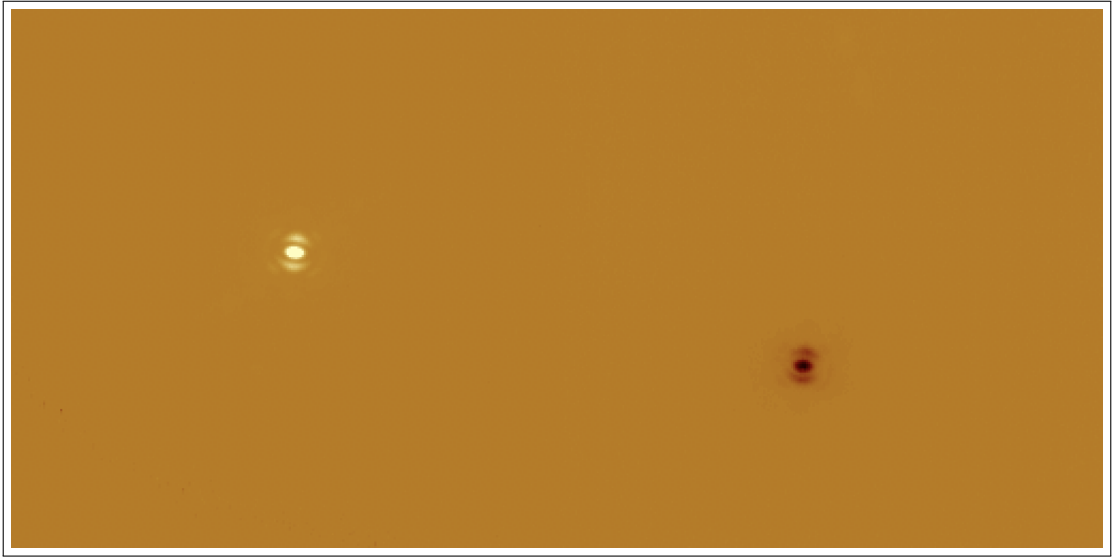


Figure 3.4: After Sky Subtraction - the target on the left (bright-yellow) has positive flux and the one on the right (black) has negative flux.

3.4 Bad Cube Removal

This step is simply to eliminate the bad cubes which are not usable for science. These cubes should have been identified and noted in the observation log, otherwise they will be removed frame by frame later in the Bad Frame Removal step.

Although there are many different situations where we may consider a cube to be bad, the simplest example would be a cube taken when the adaptive optics (AO) system fails. Bad cubes have to be removed to reduce redundant undesirable images which will not contribute to science and hence reduce processing time.

3.5 Data Cropping

This step is to crop out a subframe from the original full frame that contains interesting information (i.e. around where the positive flux target is) and at the same time removing the negative flux pixels which will be irrelevant for calibrations beyond this step.

The image is cropped about a coordinate input by the user in the code. The user can obtain the centre coordinate for cropping by looking at the image by eye and approximating the coordinate of the centre of the target. The size of the subframe is also entirely up to the user. In the case of Gamma Doradus, the size of the subframe is determined by the size of the largest possible subframe centred on the target, that does not exceed the edges of the original frame.

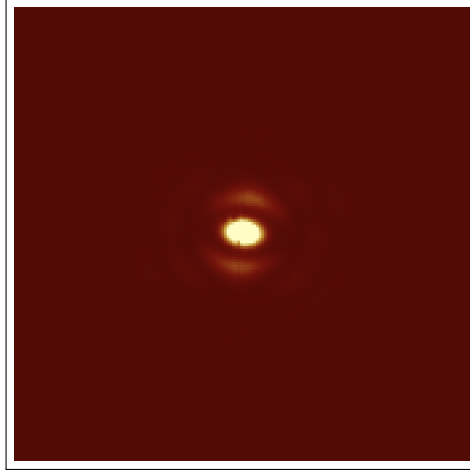


Figure 3.5: An example of a cropped image

3.6 Bad Pixel Correction

An ideal detector should have pixels of uniform sensitivity. However, that is extremely hard to achieve in reality. A real detector like the one we use in MagAO/Clio will most definitely have faulty pixels that are ultra-sensitive (hot pixels) or less-sensitive than average (dead pixels). These pixels are known as the “bad pixels”.

A function from the Vortex Imaging Processing (VIP) pipeline is used to correct for these bad pixels.^[3] The function generates a bad pixel map for each frame in each data cube via the $3\text{-}\sigma$ sigma

clipping statistics. This technique identifies a pixel as an outlier (i.e. bad pixel) if the count on the pixel deviates for more than 3σ from the median of its adjacent pixels, where the symbol σ represents the standard deviation. A sample bad pixel map is shown in Figure 3.6.

The VIP function then utilises the bad pixel map to correct for the identified bad pixels in the original frame by replacing each of them with the median value of all of their respective adjacent good pixels. According to the comments, this function supposed to work very fast with isolated (sparse pixels), however it is still the step with the longest run-time among all calibration procedures.



Figure 3.6: An example of a bad pixel map. The white spots are the identified bad pixels.

3.7 Creating Master Cube

Since we have cropped out the science target from the original frames, we no longer need to distinguish between the A1, B1, B2, and A2 science cubes, we can combine all of the science data cubes (each of 20 frames) into one single master cube. Since there are 274 science cubes in total, the master cube will contain 274×20 frames. Also note that all of these frames in the master cube have undergone linear correction, sky subtraction, subframe cropping and bad-pixel correction as elaborated previously.

Similarly for the unsaturated cubes.

3.8 Star Centring

Recall that the science target has been cropped into smaller frames about an approximate centre point of the target. However, not all images of the target is centred about that chosen point. There are a two possible methods of centring, one is by fitting the point spread functions (PSF) of the target to a Gaussian or Moffat function and another way to go is centring by Cross-Correlation.

3.8.1 Gaussian/Moffat Fitting

This centring method first requires convolution of each image in the science master cube with a 2D Gaussian function (see Figure 3.7). The full width at half maximum (FWHM) of the Gaussian function used for the convolution is determined by the estimated resolution for the instrument. To calculate the resolution, first recall the relation:

$$\theta \sim \frac{\lambda}{D} \quad (2)$$

where θ is the angle at which the first minimum of the airy disk occurs. Assuming some optical imperfections and the imperfect AO corrections, we multiply θ with a factor of 1.2, which then gives us an estimated resolution of the instrument:

$$\begin{aligned} resel &\sim 1.2 \times \frac{\lambda}{D} \text{ radians} \\ &\sim 1.2 \times \frac{\lambda}{D} \times 206265 \text{ arcsec} \end{aligned} \quad (3)$$



(a) One frame before convolution.



(b) Single frame after convolving with Gaussian function.

Figure 3.7

The Gaussian-convolved master cube is then re-centred using relevant vip functions where they fit a 2D-gaussian or moffat function to each image of the cube centred about a set coordinate. The fitting only works if the images in the master cube are not shifted drastically (i.e. $< 5px$).

3.8.2 Cross-Correlation

This is an alternative centring method. In fact, this method works better on Clio2 datasets. The vip cross-correlation function will first apply fit the centroid with a 2-D Gaussian for the first frame of the data cube, and then the remaining frames will be centred with respect to the first frame by cross-correlation where the degree of shifting of each frame will be recorded in a separate list which is also implemented in the function.

3.9 Angular Differential Imaging (ADI) and Derotation Angles

Angular Differential Imaging (ADI) is a high-contrast imaging technique that is used to reduce quasi-static speckle noise to improve the chances of detecting a nearby companion of a bright source. This technique is implemented by turning off the instrument field derotator such that the artefacts of the instrument is now static while the field of view rotates over time^[4] (see Figure 3.8 for visualisation). The artefacts can then later be subtracted and the residual images are rotated to align their field. The aligned residual images are then stacked to produce a image of significantly larger contrast.

The ADI strategy was used to collect data for Gamma Doradus using Clio2 and thus this reduction step collects information on the offset angle for each image. These offset angles will be converted into the derotation angle via the formula:

$$\text{DEROT} = \text{ROTOFF} - 180^\circ + \text{North_CLIO2} \quad (4)$$

The subtraction of 180° is to obtain an East-left image for Clio2 and $\text{North_CLIO2} = -1.8^\circ$ is to align the image to North-up.^[5,6] The derotation angle computed through this step is used for median-ADI or PCA-ADI (see later in Section 5.2 and 5.3) in post-processing, after PSF modelling and subtraction step, where the offset images are de-rotated and re-aligned to the field.

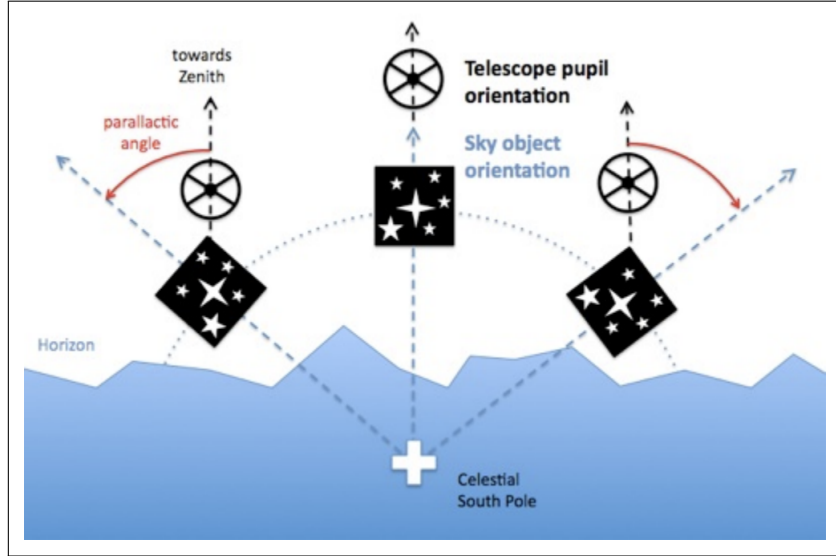


Figure 3.8: Visualisation of Angular Differential Imaging - this image demonstrates the field rotation from the point of a stationary observer by the telescope on the ground.^[7]

The rotation offset angle (**ROTOFF**) can be found in the header of the original data cubes. The **ROTOFF**'s for each individual cubes are compiled into a list. However, there are 20 image frames in each cube which are assumed to be taken over fixed time intervals. Assuming that the rotation angles in the header is assigned to the middle (10th) frame of the full 20 frames of each cube, we can extrapolate the rotation angles resulting in a list of angles which corresponds to each individual frame of the data cube. These rotation angles will be used to align the images to the field in the post-processing algorithm.

3.10 Bad Frame Removal

This final step of the data reduction procedure complements Section 3.4. Except that this step is to remove individual bad frames rather than the whole cube. These bad frames can be due to short intervals of instrumental failures which were not picked up by the observer during observation, for example when there is sudden change of observing conditions.

The purpose of this step, unlike the Bad Cube Removal, is not simply to reduce run time. Rather, it is crucial for preserving the images that are most correlated to each other while removing bad frames which may affect the quality of PSF modelling and subtraction in post-processing.

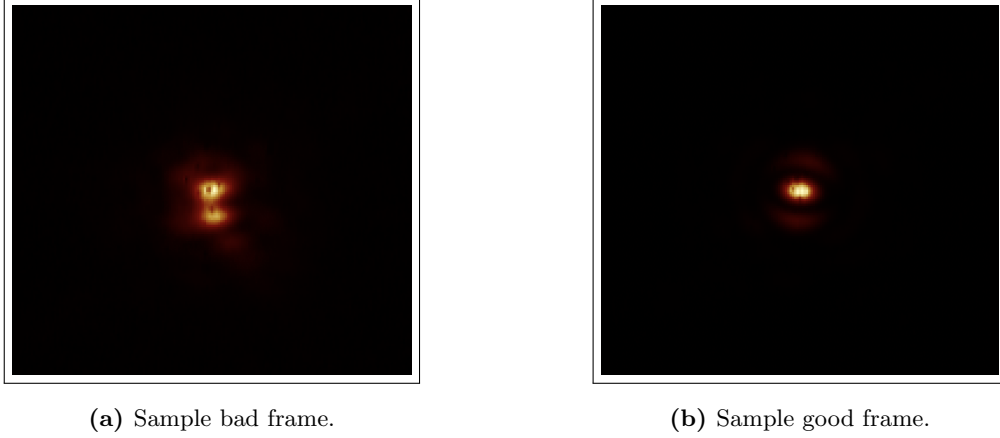


Figure 3.9

The bad frames are removed using a VIP function where the user first sets a percentage of frames that must be removed. Let's say this value is now 5% of the 5480 science frames of Gamma Doradus (i.e. 274 of 5480 images will be removed). The function will then cross-correlate each individual frame in the master cube with every other frame. The 274 frames (5% of 5480) with the smallest Pearson correlation coefficient (i.e. least correlation) with respect to other frames in the cube will then be removed.

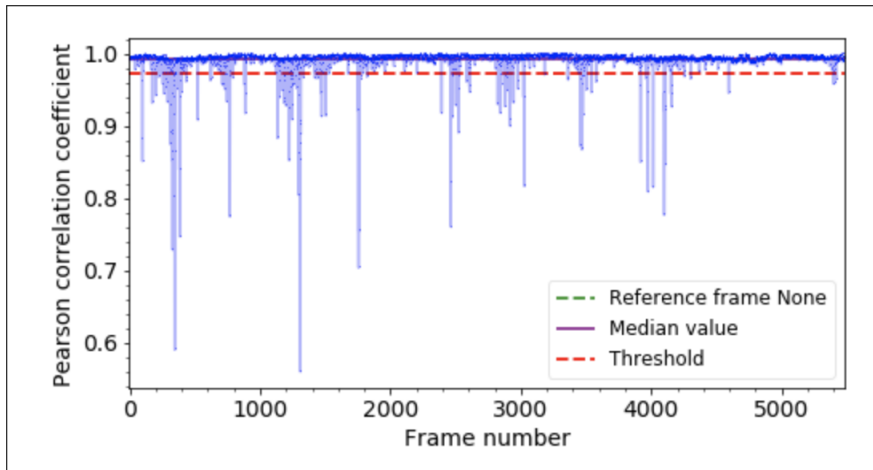


Figure 3.10: Frames with Pearson correlation coefficients below the threshold (red line) are removed.

3.11 Temporal Binning

This step is to reduce the number of frames in the data cubes. Let's say we want to reduce it by a factor of 10, we hence take the mean of every 10 frames and put the mean in a separate cube which will have 10 times less frames than the original cube. This is what we call the "binned cube".

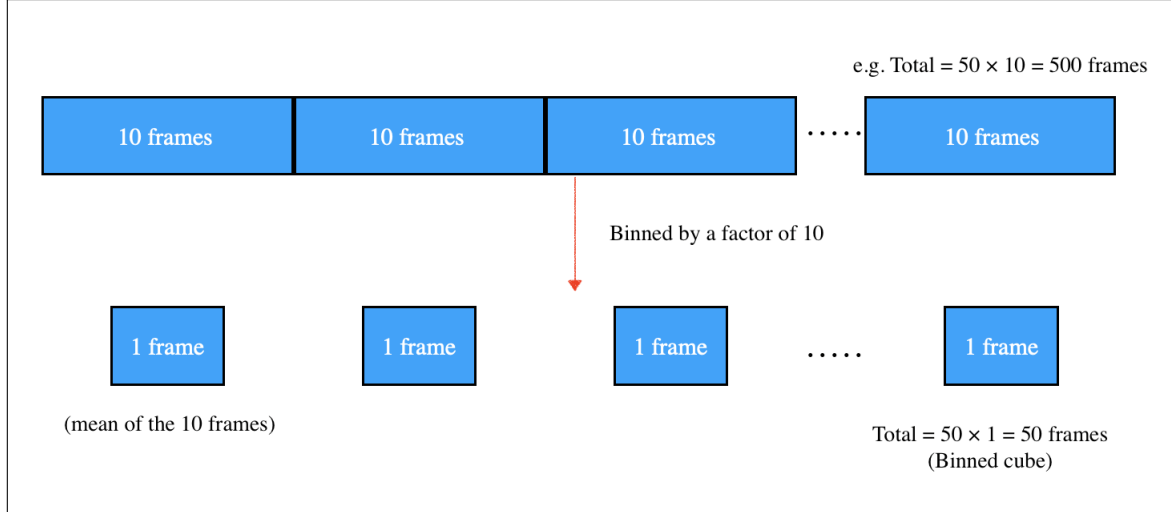


Figure 3.11: Illustration of Temporal Binning with Binning Factor = 10.

Temporal binning reduces computation time in the post-processing step, specifically for Principal Component Analysis (PCA) in concentric annuli (see Section 5.3 for detailed explanation on PCA).

4 STELLAR PHOTOMETRY

4.1 Measuring Flux for Unsaturated PSF

The flux of the saturated PSF is essential information for the post-processing step. However, flux for the saturated PSFs cannot be measured directly since it is beyond the limits of the instrument. It can only be calculated from the unsaturated flux that can be measured.

The first step to measure the unsaturated flux is to centre the unsaturated cube. In the case of Gamma Doradus, centring via Moffat fitting is the most ideal. After centring, the median of the cube is taken and a VIP function is used to fit a 2D Gaussian onto the 2D distribution of the median to obtain a FWHM in the x and y coordinate respectively. A model PSF is generated to fit the actual PSF and the FWHM is computed from the fit. The quality of the Model PSF can be determined by checking how well the centroid is subtracted in the residuals (see Figure 4.1 below).

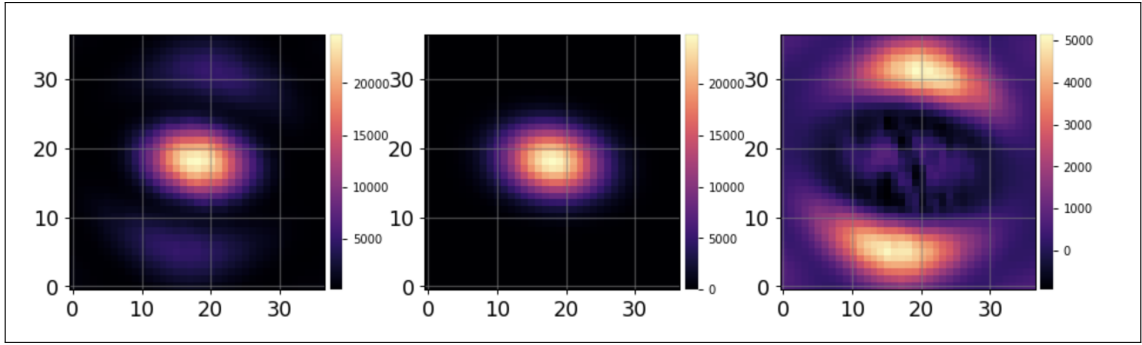


Figure 4.1: (From left to right) Actual PSF, Model PSF, Residuals after subtracting model from actual PSF.

Finally, the axis-specific FWHM are averaged to give a single-valued estimate FWHM and is used as an input parameter in another VIP function that provides the actual flux of the unsaturated median PSF while also normalising it. Normalising in this case means to scale the flux of the PSF such that the total flux in one FWHM aperture is equal to 1. This will be useful later in the post-processing step if we wish to inject a fake companion which is, for example, 5-sigma above the average noise.

4.2 Calculating Flux for Saturated PSF

Now that the flux for the unsaturated PSF is known, it can be used to calculate the flux of the saturated PSF simply using the ratio method:

$$flux_{sci} = \frac{flux_{unsat}}{DIT_{unsat}} \times DIT_{sci} \quad (5)$$

5 POST-PROCESSING

Post-processing is crucial for maximizing the contrast of the image to increase the chances of seeing a faint companion around the target. The original pipeline was written by Dr. Valentin Christiaens for NaCo data, minor modifications were required to adapt it to the MagAO data.

5.1 High Pass Filter (HPF)

This is the first step in the post-processing procedure for Thermal IR observations. [8, p. 145] Setting this option to “true” means that signals of small spatial frequencies will be filtered out, and could be useful in terms of removing large scale noise variations. However, there is a risk of missing out on extended genuine signals. Hence, it is recommended to have this option turned both on and off, then check if there are any difference between the final images (see below).

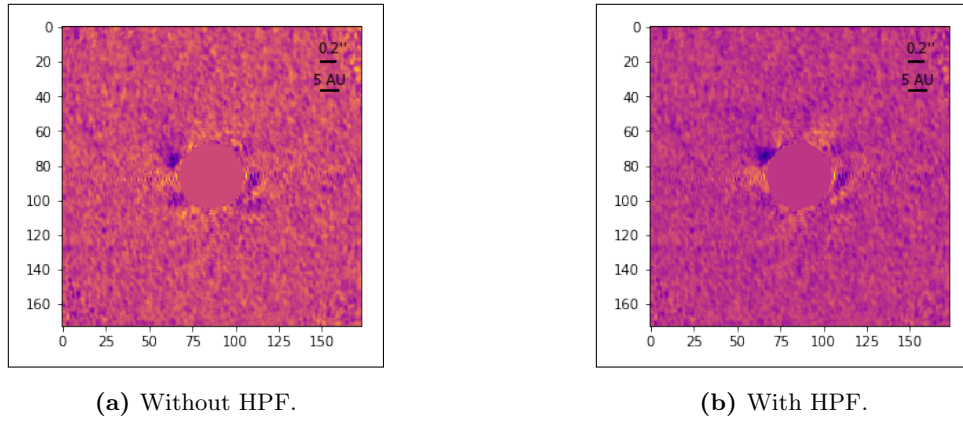


Figure 5.1: Comparison between post-processing with and without HPF. Also note that the bright source at the centre is masked so that the details around it are more visible.

5.2 Median ADI

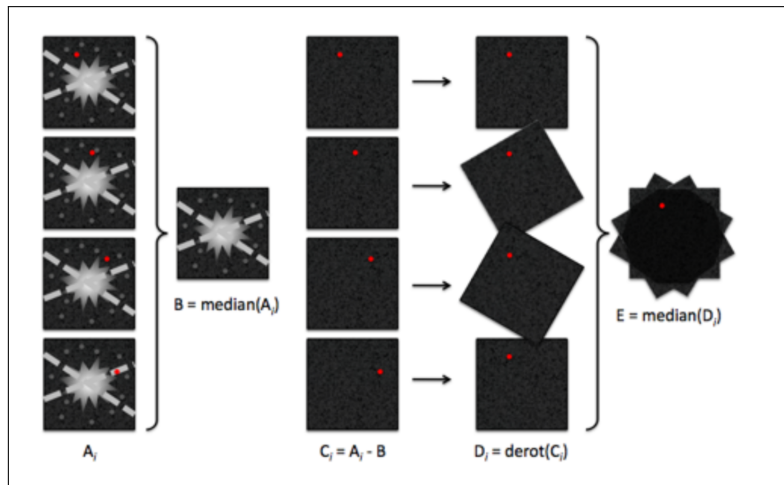


Figure 5.2: Principle of the Median ADI strategy. (credit: Thalmann [8, p.118])

There are a few different post-processing techniques implemented in this pipeline. First of which is the Median ADI technique, the simplest and least computationally intensive. This method takes the median of all frames in the reduced raw data cube to model the PSF of the star. Taking the median of all frames avoids the inclusion of the companion signal in modelling a PSF which will later be subtracted from each of the raw image. This way, any companion present in the raw image will not be removed upon PSF model subtraction. The images after PSF model subtraction are known as residual images. The residual images will undergo ADI where they are derotated and stacked. If the companion is not too faint, we might be able to see it after stacking.

5.3 Principal Component Analysis (PCA) and ADI

Principal Component Analysis (PCA) is a method used to construct a model for a specific dataset. PCA uses images in the observed datacube as a library for orthogonal principal components (basis), where the basis are ordered by decreasing variance. Here, I will run through the general overview of PCA based on Dr. Christiaens' thesis^[8, p. 123-124] :

- (1) The PCA library is a 3-dimensional cube, 2 of which are spatial and the other is temporal. The 3D cube is first mean-centred and then converted into a 2D $m \times n$ matrix with each column representing one frame in the PCA library.
- (2) The 2D matrix, S , then undergoes singular value decomposition (SVD):

$$S = UWV^T \quad (6)$$

where W is a diagonal matrix and U and V are unitary matrices. The columns of matrix V is what we are interested in. Each of the columns forms a basis, ϕ_i .

- (3) The PSF model, $R(\vec{x})$, is built upon ϕ_i for each image, $I(\vec{x})$ as such:

$$R(\vec{x}) = \sum_{i=1}^{n_{pc}} a_i \phi_i^{n_{pc}}(\vec{x}) \quad (7)$$

where a_i is the projection coefficient of $I(\vec{x})$ onto ϕ_i and the basis in matrix V is truncated such that only the first n_{pc} basis are used in PSF modelling.

- (4) The model stellar PSF be subtracted from the original image. The truncating of the basis in matrix V is hence necessary to prevent self-subtracting the whole image built upon the full set of basis, which imposes a risk of subtracting any companions that may be present.
- (5) Residual images after subtraction is then derotated (ADI) and stacked resulting in a final image of improved contrast.

The last step above is the reason why we call this technique PCA-ADI. There are two types of PCA-ADI used in the post-processing of the datacube: PCA-ADI in full frame and PCA-ADI in concentric annuli.

5.3.1 PCA-ADI in full frame

The principal components (PCs) in this technique are computed by performing SVD of the PCA library, which consists of all the observed frames in the datacube.

5.3.2 PCA-ADI in concentric annuli

PCA-ADI in concentric annuli requires the field to first be divided into concentric annuli before PCA-ADI is performed on each of the individual annulus. [8, p. 146] On top of that, the PCA library for each individual image is different. Unlike PCA-ADI in full frame which utilises the whole data cube as a library, only frames that are sufficiently far apart in time from the image we are trying to model can be used as a PC for PSF modelling. If the frames are too close in time, the angular position of the companion would not have shifted enough. Thus, if these frames were to be used as a PC to model the PSF, it again leaves us at risk of subtracting away the companion.

As a consequence, PCA-ADI in concentric annuli is computationally intensive and takes much longer to process the data. Hence it is only performed on the highest binned data with lesser frames.

5.4 Comparison between Median ADI and PCA-ADI

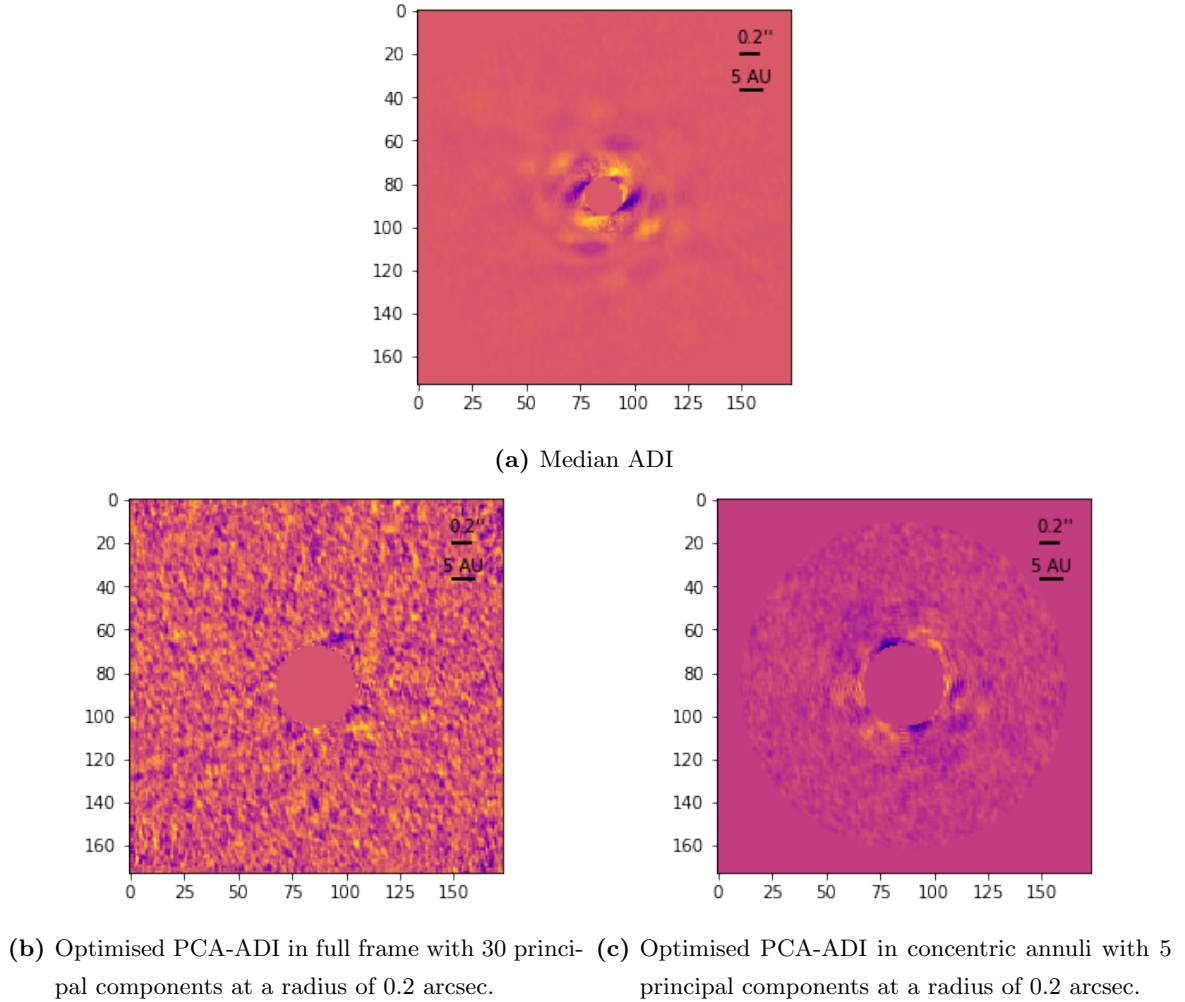


Figure 5.3: All images are without HPF. Notice that artefacts are better subtracted in PCA-ADI than the median ADI and the edges of the original square image is cut off in (c) since the field has been divided into concentric circles up to the edges of the original image.

5.5 Signal-to-Noise Ratio Map

After PSF model subtraction, we can attempt to look for companions around the host star in the residual image by measuring the significance of signals through the Signal-to-Noise Ratio (SNR) map. A companion is said to be present if the signal is above 5σ confidence level.^[8, p.124]

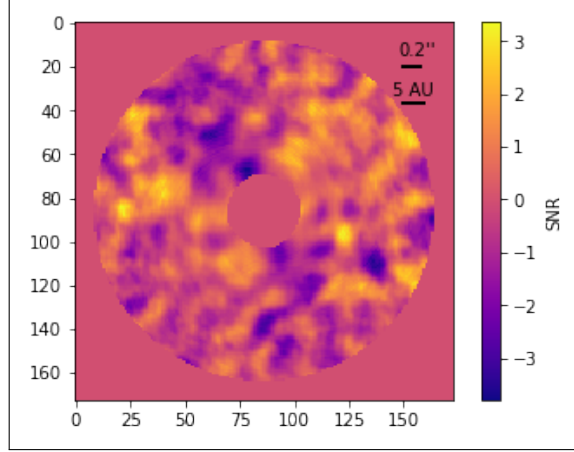


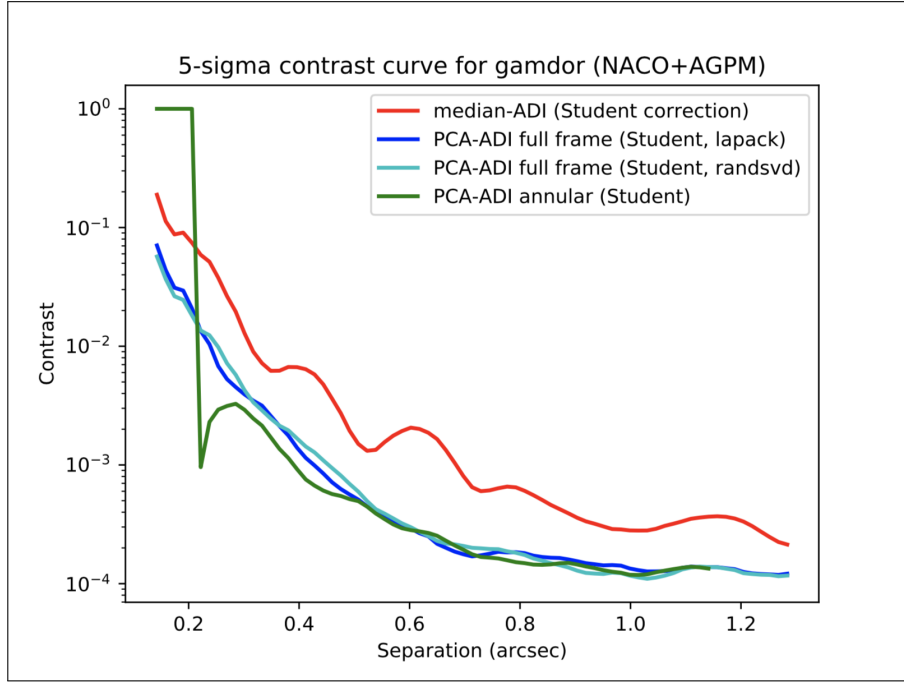
Figure 5.4: Sample SNR map. There are no regions on the image with $\text{SNR} > 5$ and hence it is safe to say that there are no companions detected in this case.

5.6 Fake Companion and Contrast Curve

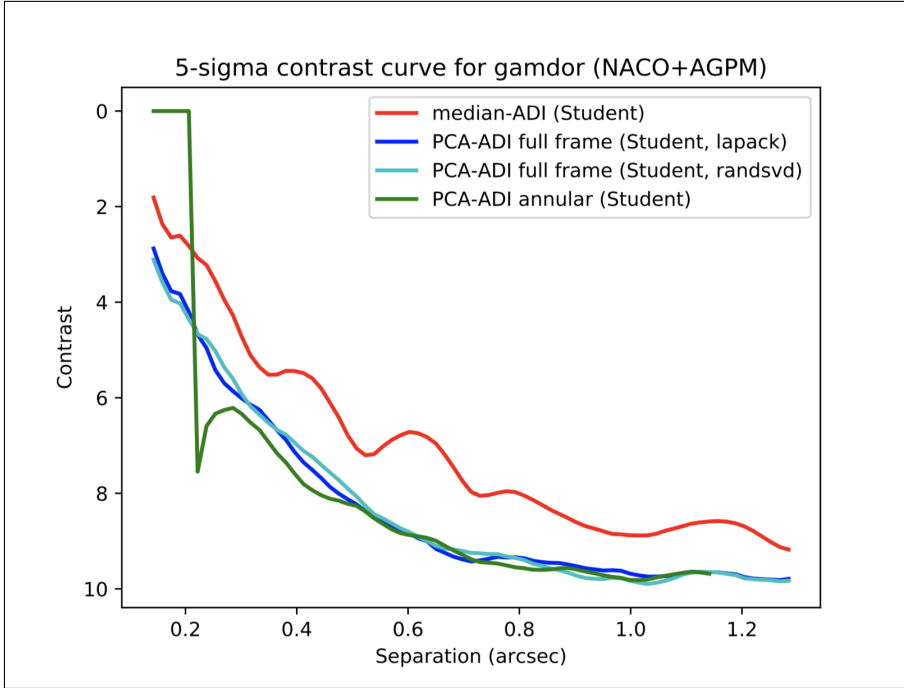
Although the post-processing of the Gamma Doradus data showed no signs of a companion, it does not exactly mean that a companion does not exist. Due to instrumental limitations, there is a contrast limit between the target and its companion above which the companion cannot be detected (higher contrast means that the companion is fainter). In other words, if the companion is too faint, we are unable to observe it.

However, it is rather possible to inject a fake companion of a known flux into the raw cube before post-processing and infer from the post-processed image, whether or not it is possible to retrieve a companion of such flux if it were to exist. This will give us a sense of how sensitive the instrument is to faint signals in the vicinity of a bright star.^[8, p. 125-126] Such information is presented in contrast curves for different post-processing methods (see Figure 5.5a).

One can probably already guess that given two companions of the same flux, the one with larger separation from the science target (host star) has a higher tendency of being detected since it is less likely to be masked by the light of the star. This is exactly what we can infer from the contrast curve where the y-axis gives the minimum flux ratio (contrast) of the companion to the host star required for detection and the x-axis gives the separation in arc-seconds between them. Hence, if we have the flux of the host star, the contrast curve allows us to determine the minimum flux of the companion required to be detected at a range of separations from the host star. The contrast curve can also be displayed on a scale of magnitude difference (see Figure 5.5b). The conversion from flux ratio to magnitude difference is explained later in Section 6.1, which then allows the contrast curve to be converted into a planet-mass sensitivity curve.



(a) Flux ratio contrast curve.



(b) Magnitude difference contrast curve.

Figure 5.5: Sample contrast curves with on different contrast scales from the post-processing of Gamma Doradus raw cube, binned temporally by a factor of 10.

6 MAGNITUDE-TO-MASS CONVERSION

6.1 Converting flux ratio into magnitude difference

The information in the contrast curve is provided in terms of fluxes and thus it is necessary to first convert that into absolute magnitudes which is compatible to the evolutionary models that can convert magnitudes of low-mass objects into masses.

The first step to take is to convert flux ratio (contrast), $\frac{f_c}{f_*}$, into magnitude difference (i.e. converting Figure 5.5a into 5.5b) via this formula:

$$M_c - M_* = -2.5 \log_{10} \frac{f_c}{f_*} \quad (8)$$

(NB: ‘c’ here represents companion.)

6.2 Variables required to obtain M_c

However, what we actually need from the flux ratio to magnitude difference conversion is the absolute magnitude of the companion, M_c . So we simply rearrange Equation 8:

$$M_c = M_* - 2.5 \log_{10} \frac{f_c}{f_*} \quad (9)$$

Hence, we need to determine the absolute magnitude of the star, M_* from two parameters: the apparent magnitude of the star, m_* and the distance to the star, d , in parsecs (pc) via the formula:

$$M_* = m_* + 5 - 5 \log_{10} d \quad (10)$$

While the value of m_* can be easily found WISE’s archive, d has to be calculated from the parallax, α , of the star which can be obtained from the GAIA archive which is given in units of milli-arcseconds (mas). To calculate the value of d in parsecs (pc), where $1pc$ is defined as the distance at which one astronomical unit (AU) stands one arcsecond (d) in the sky:

$$d = \frac{\alpha \text{ in mas}}{1000} \times \frac{1}{1AU} \quad (11)$$

(NB: small angle approximation is implemented in the calculation where $\tan(\frac{\alpha}{1000}) \approx \frac{\alpha}{1000}$.)

6.3 Evolutionary Model for Low-Mass Objects

Now that we have successfully converted the minimum contrast into the maximum absolute magnitude (faintest) required for the detection of a companion at a particular separation, we can use low-mass evolutionary models to convert these absolute magnitudes into companion masses.

The models used in this summer project are the NaCo and WISE isochrones provided by [BT Settl](#)^[9]. A sample is shown in Figure 6.1. Since the isochrone models are age-specific, we need to first have an estimate of the age of the object. The estimated age of Gamma Doradus was found to be around 0.4 – 2.0 Gyr.^[10]

The last four columns in the sample model shown in Figure 6.1 are the absolute magnitude of low-mass objects in different wavelength bands. For young planets in debris disk, emission is usually in the L-band, so we will here choose the value from W1-W10.

t (Gyr) = 0.800										
M/MsTeff(K)	L/Ls	lg(g)	R(Gcm)	D	Li	W1_W10	W2_W10	W3_W10	W4_W10	
0.0500	1443.	-4.44	5.16	6.81	0.0000	0.9290	11.924	11.615	10.457	10.410
0.0600	1716.	-4.15	5.25	6.71	0.0000	0.0000	10.965	10.866	10.085	10.072
0.0800	2436.	-3.47	5.30	7.28	0.0000	0.0000	10.109	10.010	9.566	9.529
0.0900	2656.	-3.24	5.27	7.98	0.0000	0.0000	9.637	9.466	9.154	9.074
0.1000	2812.	-3.06	5.24	8.75	0.0000	0.0000	9.256	9.068	8.818	8.727
0.2000	3260.	-2.32	5.06	15.21	0.0000	0.0000	7.666	7.488	7.338	7.246
0.3000	3415.	-1.99	4.98	20.36	0.0000	0.0000	6.911	6.753	6.621	6.534
0.4000	3521.	-1.72	4.89	26.01	0.0000	0.0000	6.291	6.157	6.034	5.953
0.5000	3680.	-1.46	4.81	32.21	0.0000	0.0000	5.689	5.605	5.488	5.422
0.6000	3968.	-1.17	4.72	38.81	0.0000	0.0000	5.090	5.084	4.970	4.935
0.7000	4386.	-0.87	4.67	44.61	0.0000	0.0000	4.588	4.643	4.541	4.528
0.8000	4823.	-0.60	4.62	50.36	0.0000	0.0025	4.199	4.248	4.171	4.161
0.9000	5216.	-0.36	4.56	57.06	0.0000	0.0788	3.834	3.871	3.813	3.806
1.0000	5586.	-0.13	4.50	64.52	0.0000	0.3060	3.487	3.514	3.471	3.465
1.1000	5910.	0.08	4.43	73.34	0.0000	0.5740	3.142	3.158	3.129	3.124
1.2000	6184.	0.27	4.36	83.67	0.0000	0.7700	2.804	2.811	2.792	2.788
1.3000	6442.	0.45	4.29	94.35	0.0000	0.8800	2.497	2.497	2.484	2.481
1.4000	6705.	0.61	4.23	104.56	0.0000	0.9380	2.229	2.226	2.216	2.213

Figure 6.1: Sample BT Settl Isochrone Model for WISE, for low mass objects with age of 0.8Gyr.

6.4 Line of best fit of isochrone model

Notice that the model provides only discrete values of masses that correspond to a certain absolute magnitude, and the absolute magnitudes that we have of the companion from the contrast curves are almost unlikely to take the values that are available in the model. The solution to that is to find the equation to the best-fit line of the model in the form of a 3rd order polynomial:

$$mass = ax^3 + bx^2 + cx + d \quad (12)$$

where a , b , c , and d are constants, and x is the absolute magnitude that we would like to convert into mass through the model. The reason why I chose a 3rd order polynomial and not any higher is because it fits sufficiently well to the model.

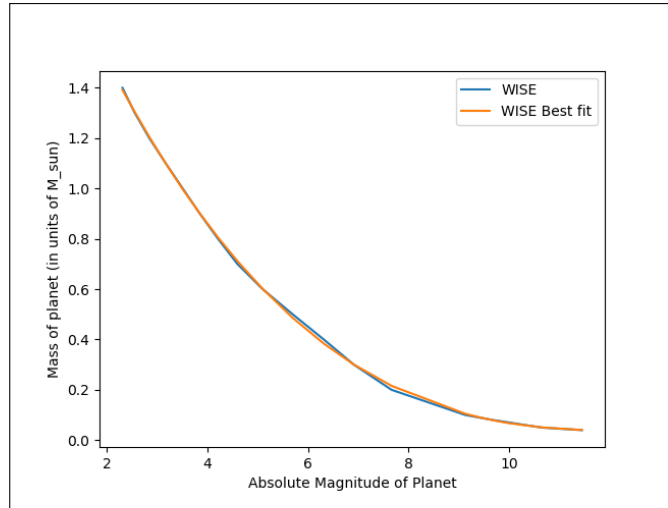
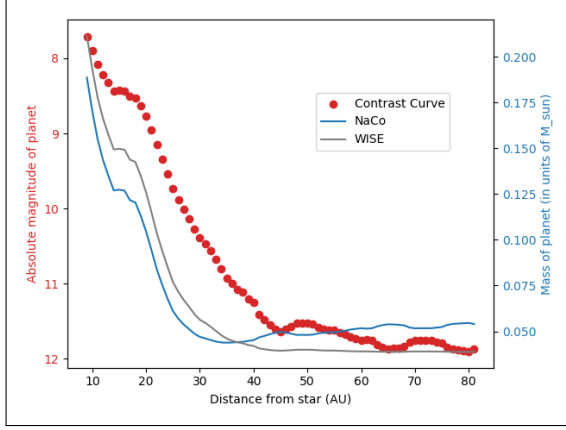


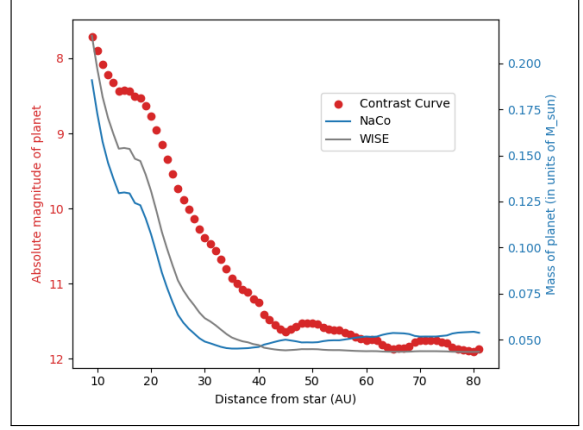
Figure 6.2: Sample best fit line vs. actual model plot to show that the best fit line fits relatively well and is usable for conversion.

6.5 Results of magnitude-to-mass conversion for Gamma Doradus

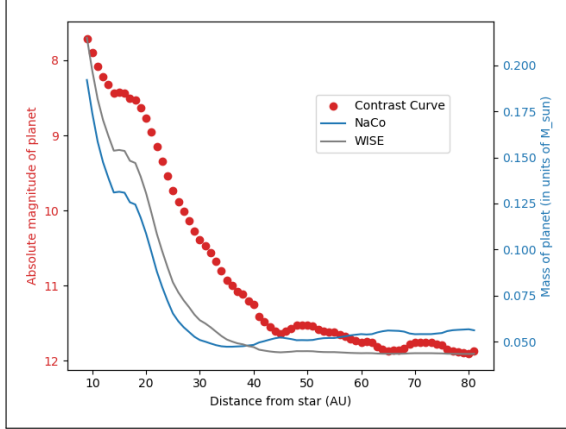
By converting the absolute magnitude of the companion into mass using the models, one can predict the mass sensitivity of an instrument for a companion at a fixed separation from the star. Note that I selected only the contrast curve from PCA-ADI full frame to avoid the “kink” present at small separations in the contrast curve of PCA-ADI annular (the green lines in Figure 5.5).



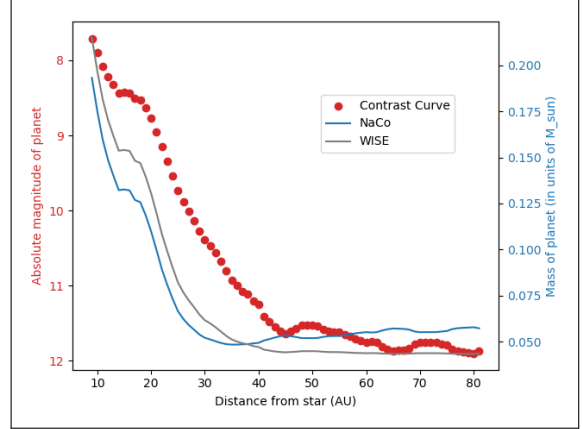
(a) Model isochrone age: 0.4 Gyr



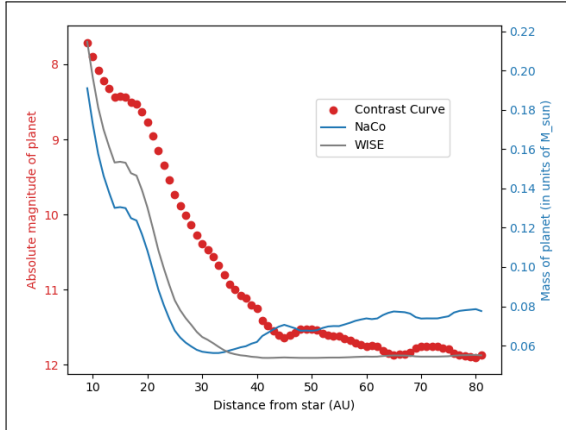
(b) Model isochrone age: 0.5 Gyr.



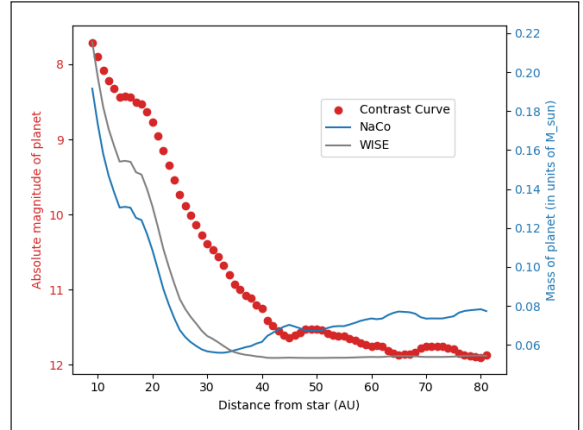
(c) Model isochrone age: 0.6 Gyr



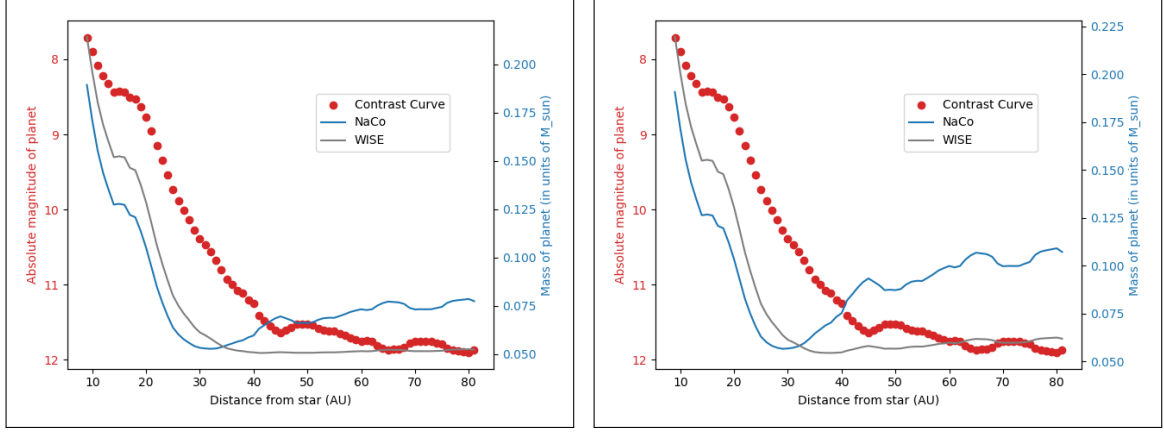
(d) Model isochrone age: 0.7 Gyr



(e) Model isochrone age: 0.8 Gyr



(f) Model isochrone age: 0.9 Gyr



(g) Model isochrone age: 1.0 Gyr

(h) Model isochrone age: 2.0 Gyr

Figure 6.3: Age-dependent magnitude-to-mass conversion of the contrast curve (red-dotted lines).

As we can see from the results, the mass sensitivity generally depends on the age of the isochrone used. Although it is more obvious in the NaCo model, both the models showed that mass sensitivity at larger separation ($> 40AU$) decreases with increasing age of the companion. This is rather logical because thermal IR emission of a planet reduces as it ages since there are no heat input after the planet has fully formed. It is simply emitting the remaining latent heat from its formation to achieve thermal equilibrium with its surrounding environment.

7 POST-PROCESSED RESULTS FOR HD35650

Two images are shown below for each of the post-processing techniques: one is the actual image of the target (HD35650), and the other one with the color bar is the SNR map which allows us to check for objects with significance of above 5σ . All images shown are the final images of post-processing without HPF.

7.1 Median ADI

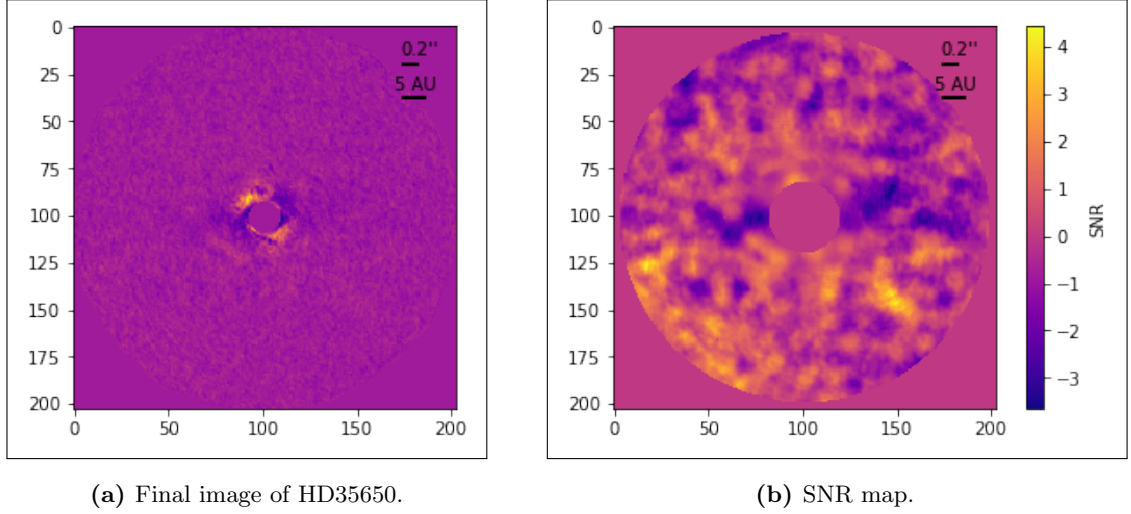


Figure 7.1: Results of performing Median ADI on the calibrated HD35650 data cube.

7.2 PCA in Full Frame

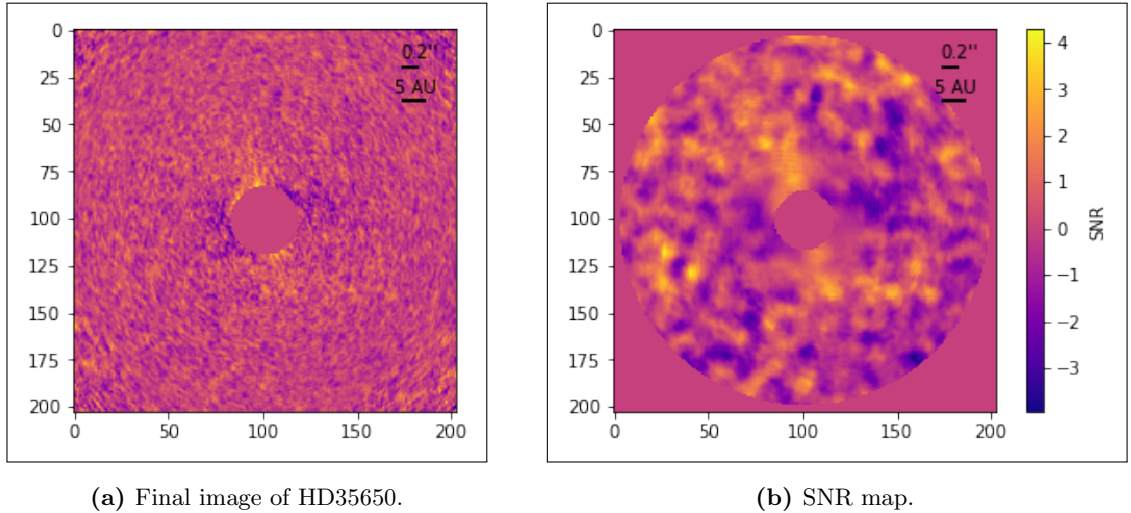


Figure 7.2: Results of optimised PCA-ADI in full frame with 5 PC at a radius of 0.2 arcsec.

7.3 PCA in Concentric Annuli

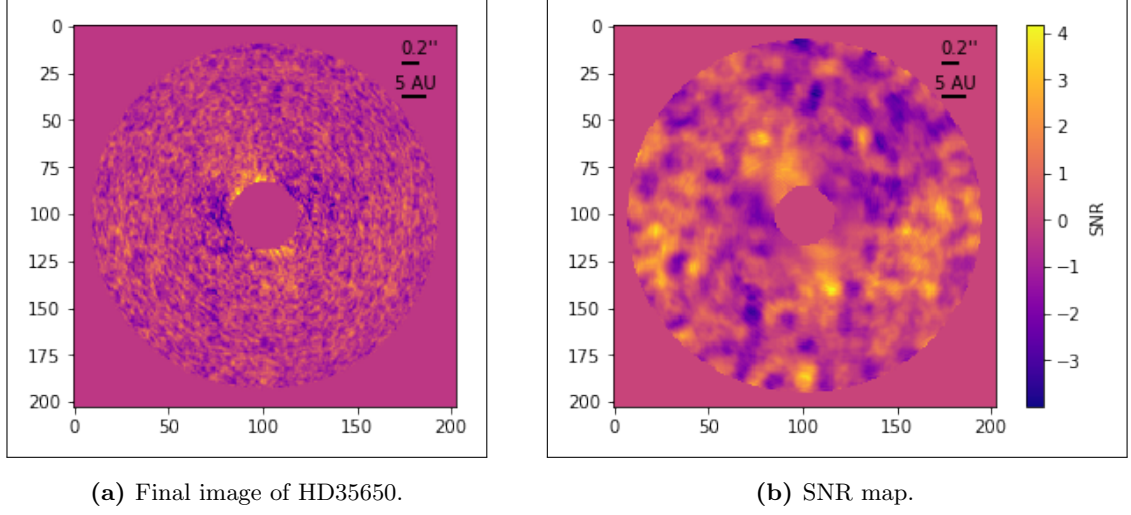


Figure 7.3: Results of optimised PCA-ADI in concentric annuli with 5 PC at a radius of 0.2 arcsec.

7.4 Contrast Curve

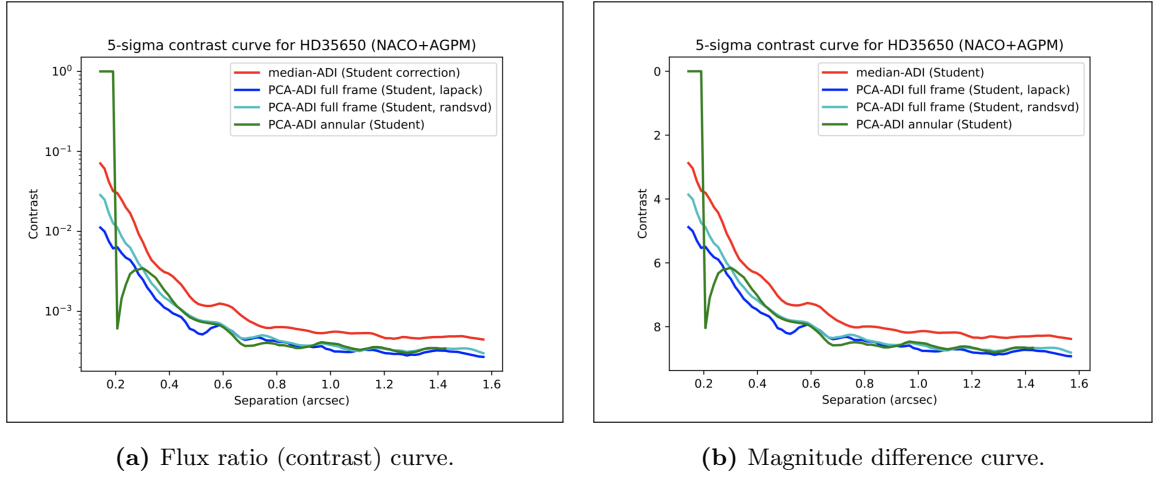


Figure 7.4: Contrast curves computed from the injection of a fake companion of known flux.

7.5 Mass sensitivity Curve

The conversion of the contrast curve into mass sensitivity curves for this data set is slightly trickier than that of Gamma Doradus. The BT Settl model did not work very well for the conversion as the magnitude range provided by the model was between 2 – 10 while the magnitude in the HD35650 contrast curve ranges from 11 – 13. Figure 7.5 shows an example of a best-fit line on the BT Settl model which we would expect to be inaccurate and unusable for magnitude-to-mass conversion beyond the magnitude ranges of the model.

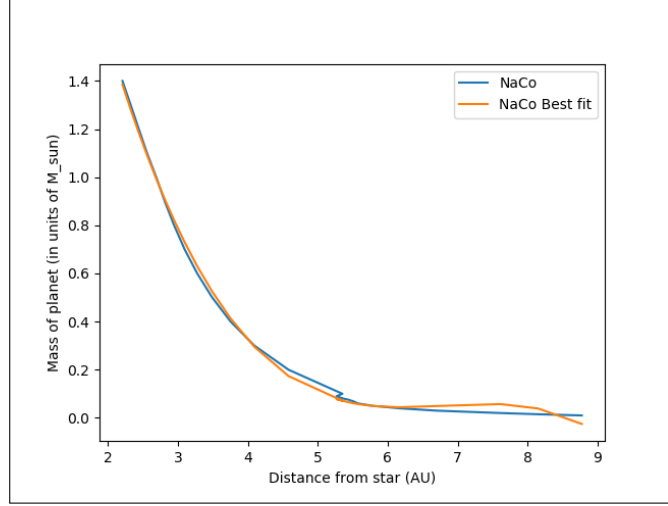
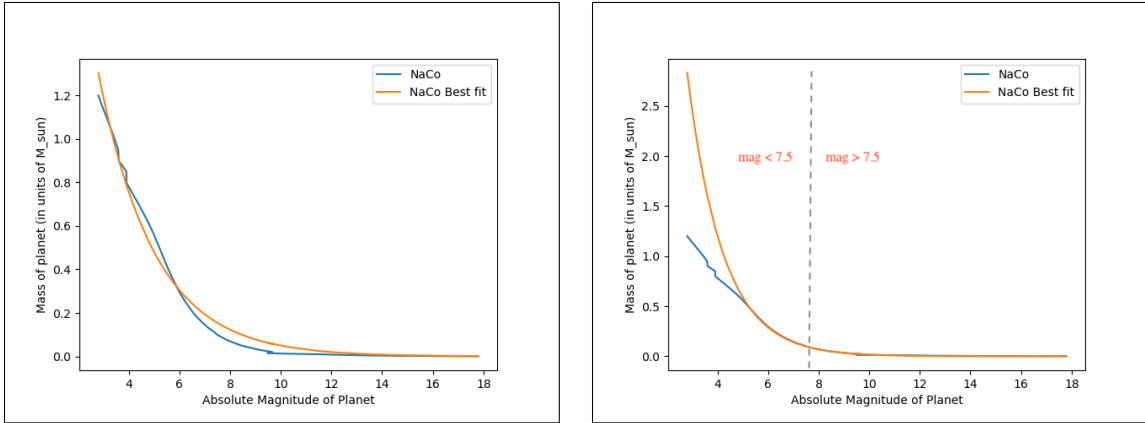


Figure 7.5: Sample best fit line vs. actual model plot. Best fit line diverges away from model pattern beyond the given ranges.

Since the BT Settl model cannot be used in this case, we turn to the AMES-Cond model, another low-mass object evolutionary model. Unlike BT Settl which is modelled upon objects with no irradiation but with cloud model^[9], AMES-Cond is an older model for objects with no irradiation nor dust opacity^[11]. Nevertheless, we still chose the AMES-Cond model for this target as the magnitude range provided by the model overlaps with the range in the contrast curve.

The AMES-Cond provides a relatively wide range of magnitude ($2 < mag < 19$) that could be converted into mass, however we are only interested in the conversion of magnitudes 10 and above into their respective masses. Hence I tried to fit a best-fit line of the model only for the magnitudes above which are relevant for the conversion. This would make more sense by referring Figure 7.6 below.



(a) Fitting a best-fit line to the whole model - not very ideal since we don't really need information on magnitudes below 7.5 (i.e. $mass \gtrsim 0.1$). (b) Fitting a best-fit line only for a selected range of magnitude/mass in the model - better fits and hence better conversion in relevant ranges.

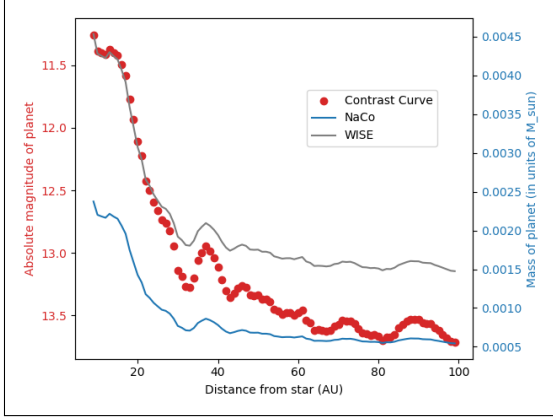
Figure 7.6: Comparison between full-model and selective best-fits.

Also, take note that the best-fit line that works best in the case of AMES-Cond is in the form of:

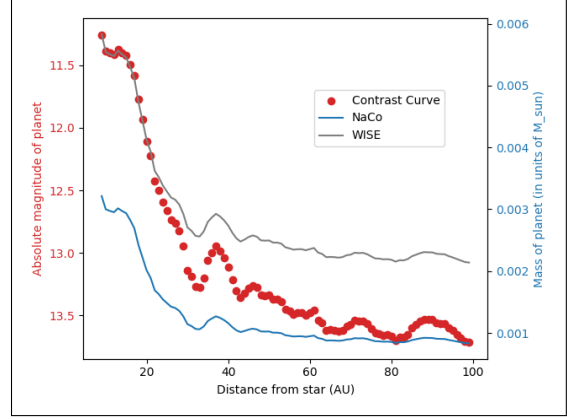
$$mass = ae^{-bx} \quad (13)$$

where again, a and b are constants, and x is the absolute magnitude to be converted into mass.

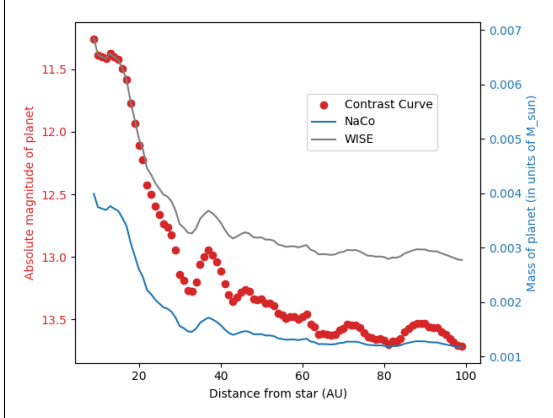
Given the age estimate of HD35650 of 4 – 250 Myr^[12], the following are the results of the magnitude-to-mass conversion.



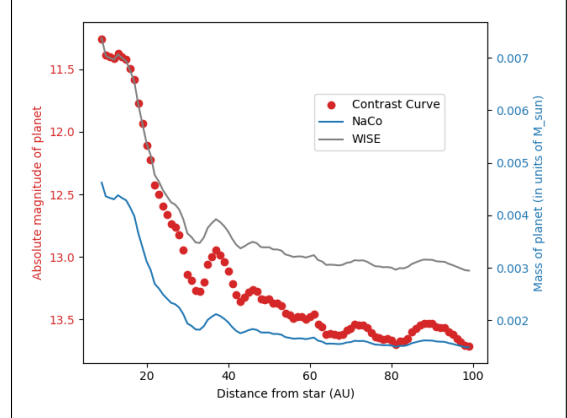
(a) Model isochrone age: 4 Myr



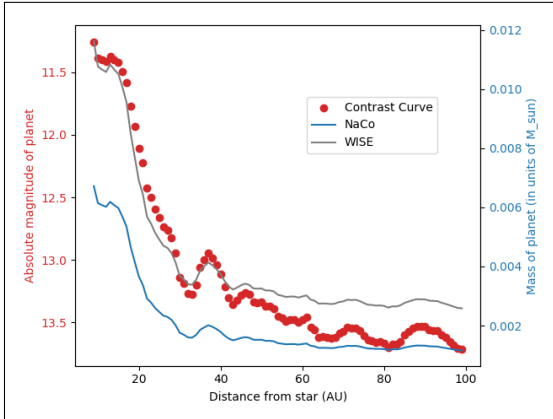
(b) Model isochrone age: 6 Myr



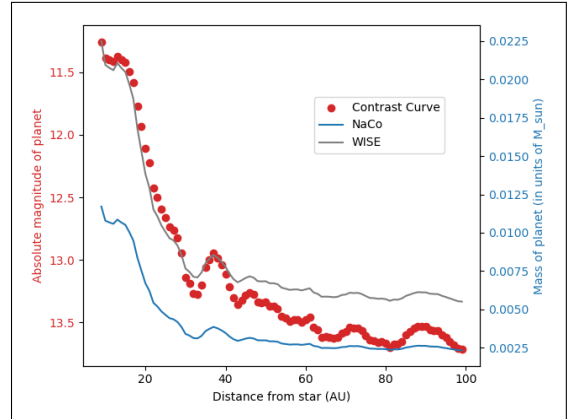
(c) Model isochrone age: 8 Myr



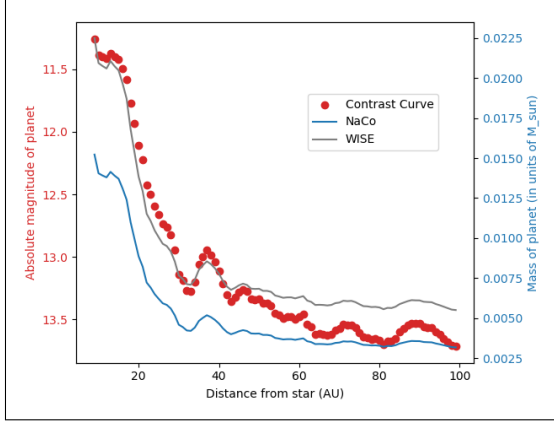
(d) Model isochrone age: 10 Myr



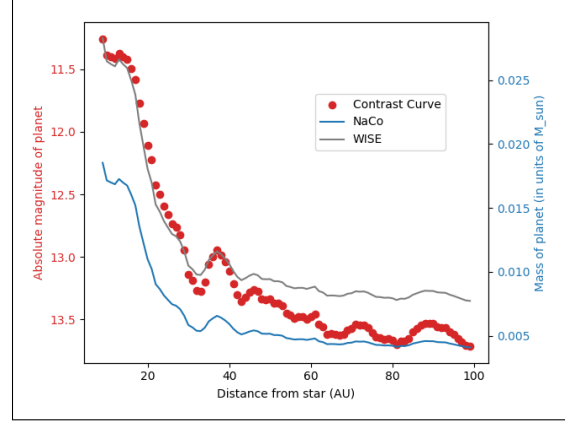
(e) Model isochrone age: 50 Myr



(f) Model isochrone age: 0.1 Gyr



(g) Model isochrone age: 0.15 Gyr



(h) Model isochrone age: 0.2 Gyr

Figure 7.7: Age-dependent magnitude-to-mass conversion of the contrast curve of HD35650.

8 SUMMARY

Although it was unfortunate that no planets have been discovered through this research project, I made use of the contrast curves from post-processing to determine the mass sensitivity of the instrument (MagAO/Clio2) for a particular companions of different separation from the host star. This information is useful in the sense that it provides future observers an idea of the chances of finding a companion around a target at a certain distance.

All in all, this summer research project was an extremely rewarding experience where I had the opportunity to understand the concepts and grasp the basic techniques of high contrast imaging - a currently growing field in planet detection. On top of that, I also learnt the techniques of data reduction and calibration which is an essential skill in astronomy and astrophysics.

9 ACKNOWLEDGEMENTS

I would like to take this opportunity to thank my supervisor, Dr. Valentin Christiaens, for his guidance and for making this research project such an amazing learning experience. Also a massive thank you to his research group who funded my project throughout the 6-week duration and to Monash University for providing undergraduates like myself with such a valuable research opportunity.

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